Pitch Level Value (PLV)

How *good* was that pitch?

How good were these pitches?

How do we measure a pitch?

Data pitches in 2020-2022, broken out into 3 feature groups:

Stuff

- Release Speed
- Horizontal Movement
- Vertical Movement
- Total Movement
- Difference from Fastball for all of the above
- Height-adjusted* Vertical Approach Angle (VAA)

Location

- Horizontal Location
- Vertical Location
- Vertical Location (relative to Hitter's strike zone)

Categorical

- Pitcher handedness
- Batter handedness
- Pitch Group
- Count

* VAAAA...AAAA

Vertical Location (relative to a Hitter’s strike zone)

Different hitters have different vertical limits for the strike zone (by design).

To account for this, we also use how far a pitch is from the center of the strike zone, in units of the vertical height of the strike zone. Pitches above the center of the zone are +SZ, and below center are -SZ.

- top of zone = 0.5 SZ
- bottom of zone = -0.5 SZ
- Distance from top to bottom = 1 SZ

You Gotta Keep 'Em Separated

Pitch types are split into 3 groups, which are then modeled separately:

Primary

- Four-Seamer
- Sinker
- Primary Cutter (if most common/only fastball used)

Breaking Balls

- Slider
- Curveball / Knuckle-Curve
- Secondary Cutter

Offspeed

- Changeup
- Splitter

* Pitch Types not listed are grouped together as "Other", and modeled separately

Baseball's Next Top Model

This process uses a combination of classification models to provide an estimate for the likelihood of the following pitch outcomes:

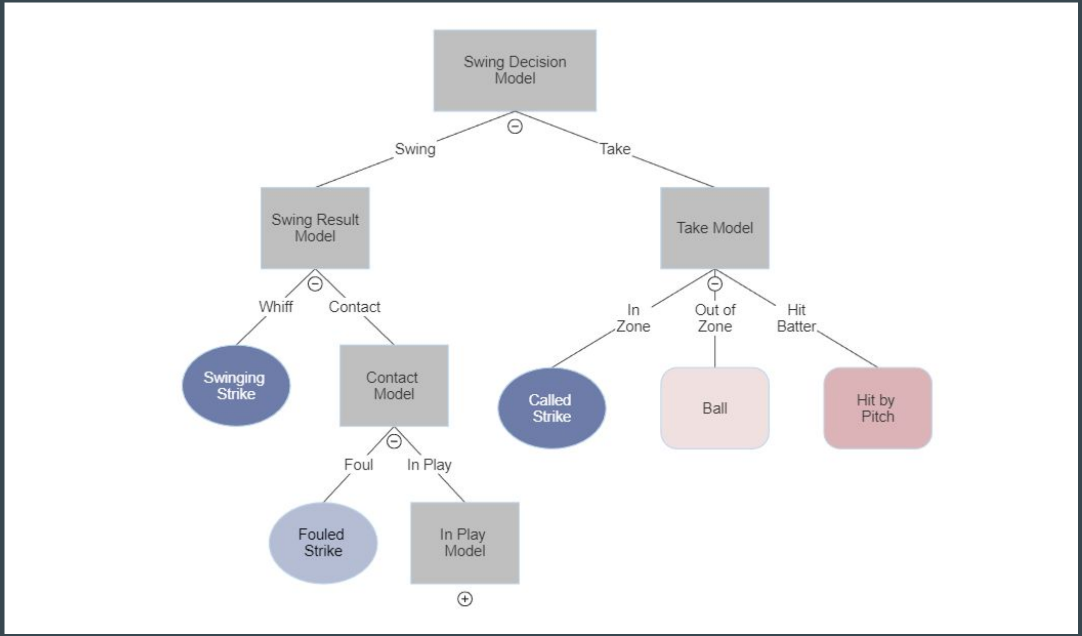
- Called Strike
- Ball
- Hit by Pitch
- Swinging Strike
- Fouled Strike
- Field Out (In-play, out)
- Single
- Double
- Triple
- Home Run

Baseball's Next Top Model (Pt 1)

Grey = Model

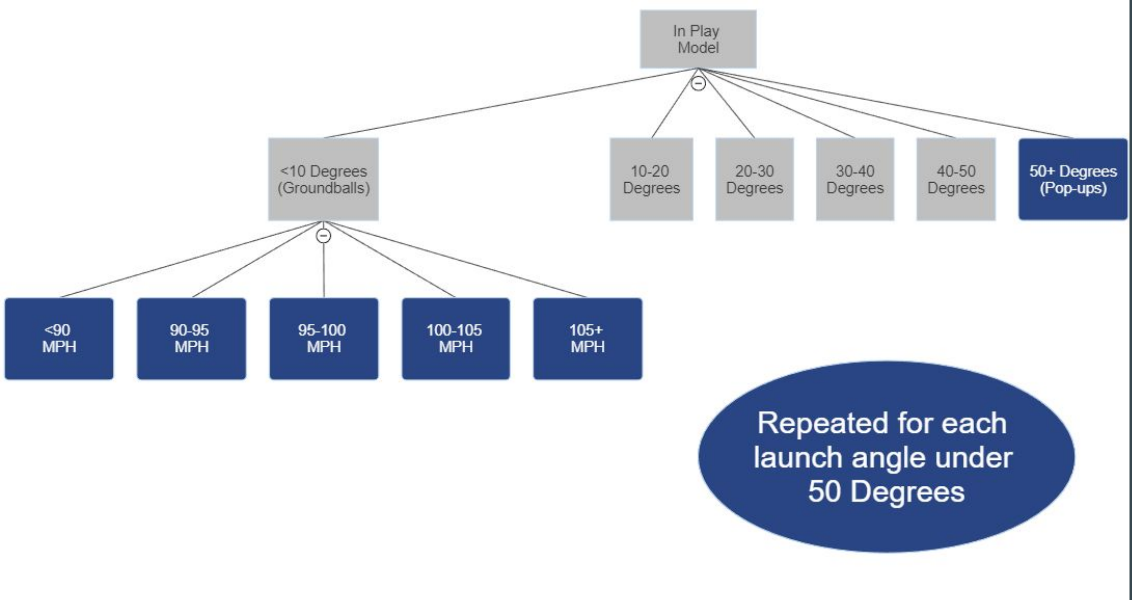
Red = Good for batter

Blue = Good for pitcher



Baseball's Next Top Model (Pt 2)

To handle balls-in-play, the likelihood of each Launch Angle (in 10 degree increments) and Exit Velocity (in 5 MPH increments) are predicted.



Baseball's Next Top Model (Pt 3)

Finally, the league-average distribution of Out/Single/Double/Triple/Home Run is applied to each Launch Angle/Exit Velocity combo.

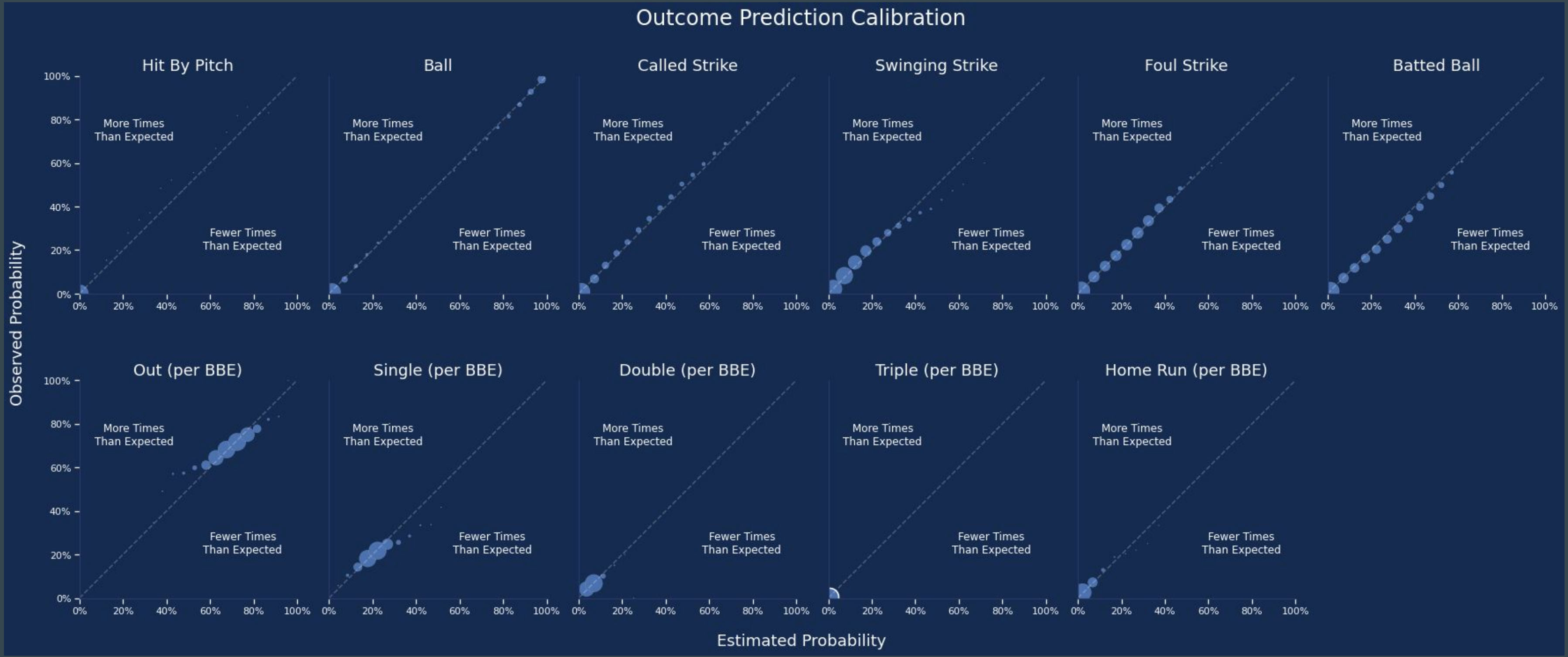
Ex. A: <10 Degrees & <90mph (weak Ground Balls)

- **Out: 83.9%**
- Single: 14.8%
- Double: 1.3%
- Triple: 0.04%
- Home Run: 0%

Ex. B: 30-40 Degrees & 105+mph (scorched Fly Balls)

- Out: 4.8%
- Single: 1.2%
- Double: 11.5%
- Triple: 1.0%
- **Home Run: 81.6%**

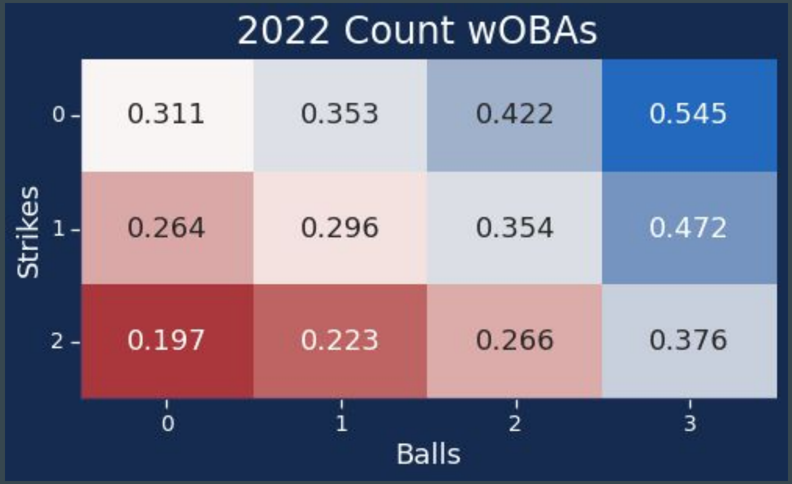
Do outcomes happen as often as we predict?



(Yes, they do!)

Valuing the Predicted Outcomes

- Each count has its own wOBA expectation (Pitchers' counts have a lower expected wOBA; Hitters' counts are higher)
- Outcome values are the wOBA value of the outcome, minus the wOBA value of the count the pitch was thrown in.
- Change in wOBA is then converted to change in Runs (ΔRuns)



One Value to Rule Them All

<u>The pitch value is the expected run value of that pitch</u>

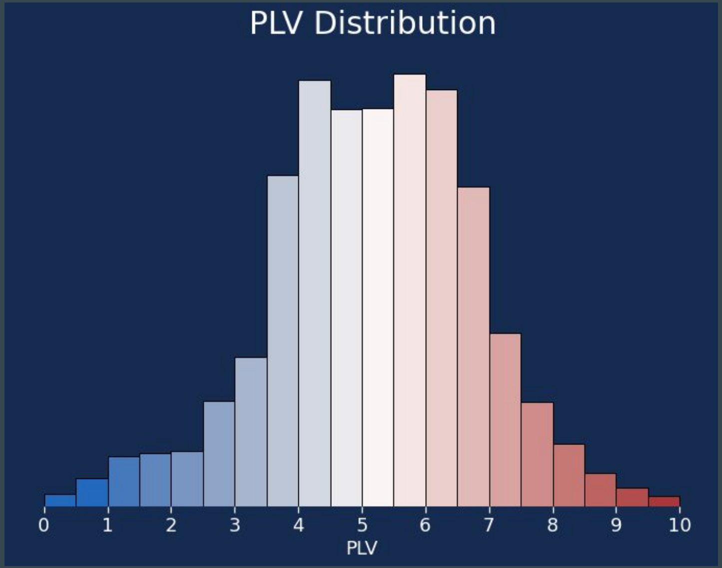
- Given a season average runs per pitch of *R*, and *n* predicted outcomes, each with likelihood *p* (%) and count-adjusted value *v* (ΔRuns), each pitch's raw value is defined as:

$$PitchValue = R + \sum_{n} p_n v_n$$

- If a pitch is expected to yield more runs than league average, that pitch was likely good for the batter
- If the pitch value is fewer runs than league average, that pitch was likely good for the pitcher

Make it intuitive

- Each pitch is compared against all other pitches in its season
- The raw values are then scaled to be 0-10, finally getting us to **PLV**
    - 10 is a great pitch (high likelihood of being a strike or out in a high leverage count)
    - 5 is always league average runs per pitch
    - 0 is a terrible pitch (high likelihood of being a ball/HBP/extra base hit in a high leverage count)
    - There are some truly valuable pitches (PLV>10) and some truly costly pitches (PLV<0), but they are *exceedingly* rare.

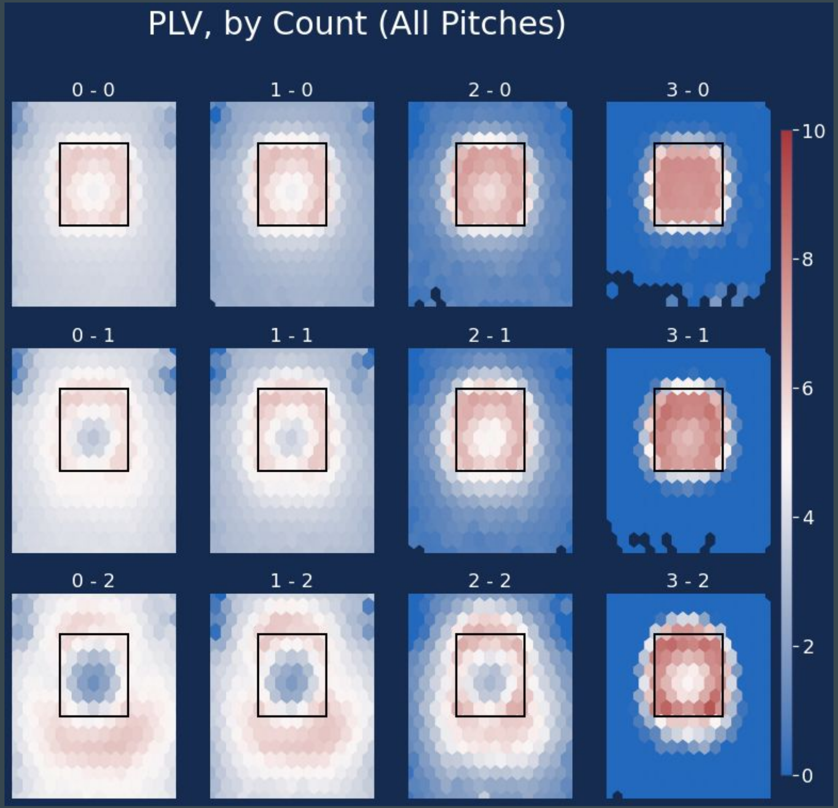


PLV Location Values, by count

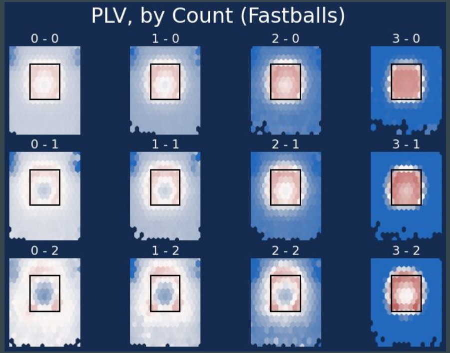
General Takeaways

Note: movement/velo also affects PLV values, but location charts are pretty.

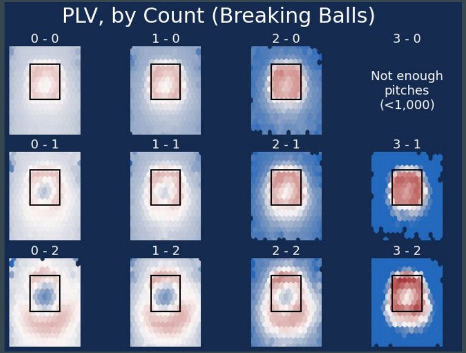
- Don't hit the hitter (<u>*very*</u> important)
- At 0-0 & 0-1, there isn't a huge cost to throwing out of zone, but there are benefits (whiff/weak contact)
- Don't throw down the middle in even/hitter's counts (#Analysis)



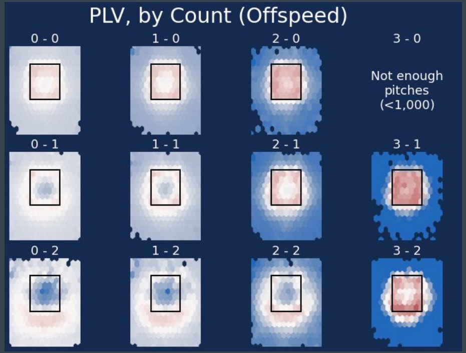
Pitch Group PLV Location Values



- High Fastballs have more margin for error than other high pitches
- Fastballs under the zone are fine, *at best*
- If you can paint the bottom corners with your Fastball, you should just do that



- Breaking Balls are the most valuable pitches in hitters' counts
  - *Especially* if those pitches are in the zone
- There's a ***lot*** of valuable real estate under the zone in 2-strike counts



- Don't hang a middle-middle Offspeed pitch (#analysis)
- In hitters' counts, Offspeed pitches have slightly more East-West margin for error than other pitch types (and a lot more in a full count)

Quality Classifier

Now that we have our PLV Values, we can start to build off of them. Our first job was to create a general classifier of pitch quality, with 3 classes:

- **Quality Pitch (QP)** : defined as any pitch with PLV >=5.5 (~44% of pitches)
- **Bad Pitch (BP)** : any pitch with PLV <4.5 (~37%)
- **Average Pitch (AP)** : any pitch between those two (~19%)

We can condense a pitcher's ability to throw Quality Pitches and avoid Bad Pitches into one metric: **QP-BP%** .

Here are 2022's top- and bottom-5 pitchers by QP-BP% (min 500 pitches):

	# Pitches	QP%	AP%	BP%	QP-BP%
Emmanuel Clase	918	62.1	17.4	20.5	41.6
Andrés Muñoz	1,015	57.6	17.3	25.0	32.6
Jacob deGrom	939	58.5	15.2	26.3	32.2
Brusdar Graterol	675	54.8	20.7	24.4	30.4
Edwin Díaz	922	56.3	14.5	29.2	27.1
Jeff Hoffman	848	38.0	17.0	45.0	-7.1
Jared Koenig	651	35.5	21.4	43.2	-7.7
Michael Pineda	731	33.7	24.4	42.0	-8.3
Justin Dunn	540	35.9	18.5	45.6	-9.6
Aníbal Sánchez	1,232	35.0	19.1	45.9	-11.0

PLA (Pitch Level Average)

- Since we now have run estimates for each pitch (the pre-PLV values), it is possible to plug those values into the ERA formula, and get an ERA-scaled value for a pitcher's overall pitch quality.
- With pitch-level estimates for each pitch type, it is possible to *also* put each pitcher's individual types on an ERA scale, using an IP proxy based on pitch usage
  - Ex: Sandy Alcantara threw 228.2 innings in 2022 and used his Changeup ~28% of the time. His Changeup's "IP" would then be 228.2 * 28% = **<u>64 "IP"</u>**
  - The PLV run estimate for all of his Changeups was 20.7 runs
  - His Changeup therefore has a PLA of: 20.7 runs * 9 / 64 IP = **<u>2.61</u>**

Hit Luck

How many of a pitcher’s hits can be attributed to luck? (and batter skill)

- Now that we have predicted likelihoods of each hit type, we can estimate how many hits a pitcher “should” have allowed, given the pitches they threw.
- The same is true for batters: the pitches they saw have a certain likelihood of being hits. How often did they capitalize, and turn those pitches into actual hits?

Hitter Performance

How well did a hitter perform, given the quality of pitches they faced?

We can apply the same pitch-level analysis, and give credit to the batter.

- Do they take a pitch when they should?
- Are they selectively swinging at pitches that they can do damage to, or are they swinging freely?
- When they make contact, how challenging was that pitch to hit?
- If they make contact, how many bases did they add/subtract, compared to the expectations of that pitch?

High Whiff - SwStrike Prob 83.6%

https://baseballsavant.mlb.com/sporty-videos?playId=9ec78ee5-0915-4b29-bcba-adf495a0c429

Unlikeliest Swinging Strike - SwStrike Prob 0.7%

https://baseballsavant.mlb.com/sporty-videos?playId=aa286d2d-da85-4188-ad2b-c37eaca863c9

Painted Corner - Called Strike Prob 85.3%

https://baseballsavant.mlb.com/sporty-videos?playId=6fdf37b4-988c-4c9f-960e-551ca6c0b415

The Meatball - Home Run Prob 35.9%

https://baseballsavant.mlb.com/sporty-videos?playId=1b7e21c3-5daf-4d2e-9922-ff1a82fa4fae

Biggest Miss - Home Run Prob 16.6%

https://baseballsavant.mlb.com/sporty-videos?playId=1999bf4e-d646-428b-b0b8-85047c12a7d9

Extensions of PLV

- Separate it into components
  - Physical characteristics (aka Stuff)
  - Location
- Hitter attributes
  - Swing Decisions
  - Contact over expected
  - Power over expected
- Catcher Framing (Called Strikes above expected)
- Catcher's ability to call a game
  - Are they having their pitcher throw their best pitches?
  - Do they notice when a pitcher doesn't have a pitch that day, and call it less?

PLV 2.0... and beyond!

Ideas for new features:

- More spin information!
- Seam Shifted Wake
- Tunneling
- Arm slot estimation
- Sequencing
- Using rest of the pitch mix
    - How do pitches play off of each other
- Adding spray angle predictions? (Pull/Center/Oppo)